

Publicly Documented Mirror Specifications for ASML/ZEISS EUV Lithography Systems

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1 Purpose, terminology, and limits of public information

EUV scanner optics are frequently described using a mixture of official product specifications, historic development hardware, patent examples, and laboratory multilayer samples. These sources are all useful, but they do not have the same evidentiary meaning. This article therefore uses the following labels:

- **Official**: information stated by ASML or ZEISS for an actual scanner platform or optical subsystem;
- **Paper**: a measured or designed result reported in a scientific publication or institutional technical document;
- **Patent**: a numerically complete Carl Zeiss embodiment that is relevant to real EUV optical design, but is not identified as the prescription of a current production scanner;
- **Not disclosed**: a quantity for which no authoritative public production value was located.

The word *mirror* is also used at several levels. The collector is a source optic; the field- and pupil-facet assemblies contain many independently directed reflective facets; the mask itself is a patterned reflective multilayer; and the projection objective contains the imaging mirrors that form the reduced image on the wafer.

2 Optical architecture of an EUV scanner

The functional light path is

$$\begin{array}{c} \text{Sn plasma} \longrightarrow \text{collector} \longrightarrow \text{intermediate focus} \longrightarrow \text{illuminator} \\ \longrightarrow \text{reticle} \longrightarrow \text{projection optics} \longrightarrow \text{wafer.} \end{array} \quad (1)$$

The source-side optics collect, spectrally select, homogenize, and angularly distribute the radiation. The projection objective then accepts the diffraction orders from the patterned reflective mask and forms a reduced image at the wafer. The entire EUV optical path operates in vacuum because air and ordinary transmissive optical materials strongly absorb radiation near 13.5 nm [1, 6].

3 Officially disclosed system-level specifications

Table 2 contains quantities that can be stated for actual ASML platforms without borrowing a patent prescription. ASML states that the NXE:3600D uses a 13.5 nm source, reflective projection optics with $\text{NA} = 0.33$, and a maximum $26 \times 33 \text{ mm}^2$ exposure field [2]. The EXE:5000 product description gives $\text{NA} = 0.55$, 8 nm resolution, anamorphic projection optics, and an exposure field

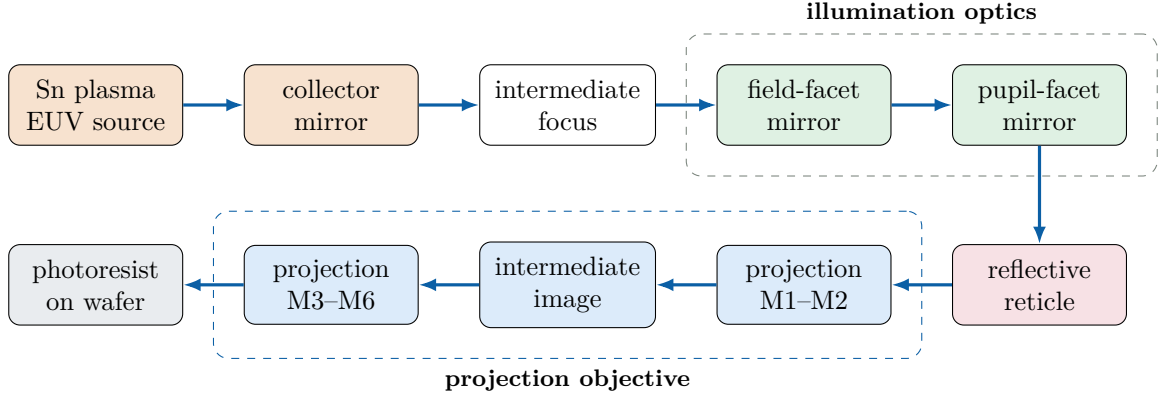


Figure 1: Functional EUV optical chain. The grouping of M1–M2 and M3–M6 follows the detailed patent embodiment in Sec. 5; this is a functional schematic, not the folded geometry or proprietary prescription of a production scanner.

Table 1: Principal reflective elements and their publicly documented functions.

Element	Main optical function	Publicly documented structural information	Evidence class
Collector	Captures a large solid angle from the Sn plasma and forms the source at the intermediate focus	Approximately ellipsoidal normal-incidence optic with a central laser aperture and a laterally graded multilayer in published Cymer source hardware	Paper
Field-facet mirror	Divides and redistributes source power into illumination channels; contributes to field/slit uniformity at the mask	Many reflective facets; High-NA uses actuated field-facet technology	Paper
Pupil-facet mirror	Maps illumination channels into the desired angular distribution at the reticle	Enables conventional, annular, dipole, quadrupole, and source-mask-optimized pupils	Paper
Reflective reticle	Encodes the circuit pattern and diffracts the incident EUV field	Multilayer reflector plus patterned absorber; not part of the six imaging mirrors	Official/Paper
Projection M1–M6	Forms a reduced, aberration-controlled image of the reticle on the wafer	Six imaging mirrors are documented for standard-NA optics; detailed modern surface data remain proprietary	Official/Patent

Table 2: Public system-level data for representative standard-NA and High-NA ASML platforms.

Property	NXE standard-NA EUV (e.g. NXE:3600D)	EXE High-NA EUV (e.g. EXE:5000)
Nominal EUV wavelength	13.5 nm	13.5 nm
Image-side numerical aperture	0.33	0.55
Published resolution figure	approximately 13 nm platform class	8 nm
Maximum wafer exposure field	$26 \times 33 \text{ mm}^2$	$26 \times 16.5 \text{ mm}^2$ half field
Projection principle	reflective, nominally $4\times$ reduction	reflective and anamorphic; commonly reported as $4 \times / 8\times$ reduction in the two orthogonal directions
Projection mirror architecture	six imaging mirrors for the established ZEISS Starlith architecture	larger, more extreme aspheres; last mirror has a hole producing a central obscuration
Illuminator	field-facet/pupil-facet architecture	optimized three-mirror illuminator using actuated field facets
Exact modern surface prescription	Not disclosed	Not disclosed
Exact production multilayer recipe	Not disclosed	Not disclosed

half the area of the NXE field [3]. The ZEISS/ASML optical-development paper gives the half-field dimensions as $26 \times 16.5 \text{ mm}^2$ and describes the High-NA optical architecture [6].

ASML further states that each EUV mirror contains more than 100 engineered material layers and is polished to a smoothness below roughly one atomic thickness [1]. This means “more than 100 layers” should be interpreted as material layers, not necessarily more than 100 *bilayer periods*. A 50-period Mo/Si stack already contains 100 constituent layers before including interlayers and a cap.

ZEISS reports that High-NA mirrors are significantly larger than standard-NA mirrors and that High-NA projection optics and illumination optics require new fabrication, metrology, and mechatronic infrastructure [5, 6]. These official descriptions establish the scale and precision class, but do not supply the surface-by-surface production prescription.

4 Source-side collector and illumination mirrors

4.1 Collector mirror

In a laser-produced-plasma (LPP) source, pulsed laser energy is delivered to a tin target and generates a broadband plasma emission centered around the useful 13.5 nm band. A published Cymer source developed to ASML requirements used an approximately ellipsoidal normal-incidence collector with the following reported characteristics [8]:

- collector diameter greater than approximately 650 mm;
- collection solid angle greater than approximately 5 sr;
- plasma located approximately 200 mm from the collector;
- a central aperture for the incident CO₂ laser beam;
- a graded multilayer coating to accommodate the spatial variation of incidence angle.

These numbers describe a published HVM-source development generation, not a dimensional drawing of the latest source module. Their optical rationale is nevertheless direct. If the source

plasma is placed near one focus of an ideal ellipsoid, specular reflection directs paraxial rays toward its second focus, which becomes the scanner intermediate focus. The large solid angle improves collection efficiency, while the central aperture allows the drive laser to reach the tin target.

The local incidence angle changes strongly over a large, curved collector. Therefore, its multilayer period cannot generally be treated as a single constant; one requires a lateral prescription

$$d = d(x, y), \quad (2)$$

chosen so that the local reflectance band remains aligned with the useful source band.

4.2 Field-facet and pupil-facet mirrors

The illuminator is not a single focusing mirror. It must simultaneously produce a uniform intensity distribution over the scanning slit and a prescribed angular spectrum at the mask. A field-facet mirror divides the source into channels and redistributes those channels across the reticle field. A pupil-facet mirror sends the channels into selected regions of the illumination pupil. In this way, the illuminator synthesizes source shapes such as conventional, annular, dipole, quadrupole, and free-form source-mask-optimized pupils.

For High-NA optics, Wischmeier *et al.* report that optimizing mirror count and column transmission led to a three-mirror illuminator based on actuated field facets. The design was evaluated at a 20 % pupil-fill ratio and supports free-form source-mask-optimization pupils [6]. The publicly stated “three-mirror” count refers to the optical architecture; it should not be confused with the number of individually actuated facets.

5 A complete six-mirror Zeiss patent prescription

The exact current Starlith surface prescription is not public. Carl Zeiss SMT patents, however, contain complete numerical examples that show the design scale, incidence-angle management, and aberration strategy of genuine six-mirror EUV objectives. The preferred embodiment reproduced below is from US 2007/0070322 A1, *Projection System for EUV Lithography*, assigned to Carl Zeiss SMT AG [7].

Correct interpretation

The following values are exact for the cited *patent embodiment*. They are not the manufacturing values of an NXE:3600D or another current scanner. The embodiment operates at 13.4 nm and NA = 0.22, whereas modern NXE optics operate at 13.5 nm and NA = 0.33.

5.1 Power sequence and optical grouping

From mask to wafer, the design uses

$$M_1(P) \ M_2(N) \ M_3(P) \ M_4(P) \ M_5(N) \ M_6(P), \quad (3)$$

or the sequence $PNPPNP$, where P denotes a concave surface with positive optical power and N a convex surface with negative optical power. Mirrors M_1 and M_2 form a first group with magnification approximately -1.02 , producing a physically accessible intermediate image between M_2 and M_3 . Mirrors M_3 – M_6 form a second group with magnification approximately -0.25 . The overall magnification is therefore approximately $+0.25$, corresponding to $4\times$ reduction [7].

The patent explains that the distribution of powers, the stop location, and the aspheric terms balance third-order astigmatism and coma between the two groups. The last mirror forms a telecentric image: the chief rays at the wafer are parallel to the optical axis.

Table 3: Performance summary of the preferred Zeiss patent embodiment [7].

Metric	Patent value
Wavelength	13.4 nm
Image-side numerical aperture	0.22
Ring-field radius at image	30.0 mm
Ring-field width at image	2.0 mm
Ring-field chord at image	26.0 mm
Nominal reduction	4:1
Overall object-to-image length	1500 mm
Composite RMS wavefront error	$0.011\lambda \approx 0.147 \text{ nm}$
Reported field variation of RMS WFE	0.007λ – 0.014λ
Chief-ray distortion	$< 1.5 \text{ nm}$
Image-space telecentricity	exit pupil at infinity

5.2 System performance of the preferred embodiment

5.3 Mirror radii, incidence angles, and aspheric departures

The signed radius follows the patent’s optical sign convention. Concave/convex classification is therefore listed independently and is the safer physical descriptor when importing the data into software with a different sign convention.

Table 4: Mirror-by-mirror values for the preferred Zeiss patent embodiment [7]. AOI is measured from the local surface normal for the chief ray from the central field point.

Mirror	Form	Vertex radius R (mm)	Chief-ray AOI (deg)	Maximum aspheric departure (μm)
M_1	concave	-643.5439	9.9	36.0
M_2	convex	-2742.4833	14.6	2.5
M_3	concave	-2850.8680	11.0	9.4
M_4	concave	1471.8440	4.4	14.0
M_5	convex	252.1454	11.0	2.7
M_6	concave	558.1165	3.1	4.0

The incidence angles are deliberately held between 3.1° and 14.6° . This reduces polarization-dependent amplitude variation and multilayer-induced phase error, and reduces the severity of lateral period grading. The maximum departure from a best-fit sphere ranges from $2.5 \mu\text{m}$ to $36 \mu\text{m}$. These departures are enormous compared with the wavelength, but they describe the nominal aspheric figure, not the residual fabrication error.

5.4 Aspheric sag equation and exact coefficients

Every mirror in this embodiment is a rotationally symmetric conic with even-order polynomial deformation. The sag is

$$z(h) = \frac{ch^2}{1 + \sqrt{1 - (1 + K)c^2h^2}} + Ah^4 + Bh^6 + Ch^8 + Dh^{10} + Eh^{12}, \quad c = \frac{1}{R}, \quad (4)$$

where h is the radial coordinate, K is the conic constant, and A through E are polynomial deformation coefficients. If z, h, R are entered in millimetres, the coefficient units are $A : \text{mm}^{-3}$, $B : \text{mm}^{-5}$, $C : \text{mm}^{-7}$, $D : \text{mm}^{-9}$, and $E : \text{mm}^{-11}$.

Table 5: Exact aspheric coefficients from Table 2 of US 2007/0070322 A1 [7]. All coefficients use millimetres in Eq. (4).

Mirror	K	A (mm ⁻³)	B (mm ⁻⁵)	C (mm ⁻⁷)	D (mm ⁻⁹)	E (mm ⁻¹¹)
M_1	-1.0232	2.3506e-9	-5.2889e-14	1.1199e-18	-2.2037e-23	4.1964e-28
M_2	-207.7484	1.0280e-9	2.5446e-14	6.3412e-18	-2.2982e-21	3.2649e-25
M_3	-3.9282	1.0123e-10	8.1343e-17	-1.5369e-20	3.5192e-25	-3.4955e-30
M_4	0.1886	1.9835e-11	-1.8278e-16	2.1374e-21	-1.4503e-26	5.3855e-32
M_5	0.9822	-1.3413e-8	9.1993e-13	-2.2661e-17	-5.9605e-22	2.6224e-25
M_6	0.0814	-2.4749e-11	-5.9262e-17	-1.2870e-23	-1.4070e-26	6.3467e-31

5.5 Coating-induced phase compensation

The coating is part of the optical design, not a passive “paint” applied after the geometric design is complete. For a uniform 40-bilayer Mo/Si coating with a 7.04 nm bilayer thickness on M_2 and M_5 , the patent reports the phase errors

$$\Delta\phi_{M_2} = -0.406\lambda, \quad (5)$$

$$\Delta\phi_{M_5} = +0.484\lambda, \quad (6)$$

$$\Delta\phi_{M_2+M_5} = +0.078\lambda \approx 1.05 \text{ nm}. \quad (7)$$

The opposite signs arise because upper and lower pupil rays exchange their relative positions after the intermediate image. Pairing the mirrors so that their incidence-angle variations oppose one another substantially cancels the multilayer-induced wavefront error [7]. The nominal thickness of 40 such periods is

$$40 \times 7.04 \text{ nm} = 281.6 \text{ nm}, \quad (8)$$

before a cap and any distinction between deposited and reacted interface thicknesses.

6 Mo/Si multilayer structure and optical physics

6.1 Why a multilayer is necessary

At 13.5 nm, the complex refractive index of condensed matter is close to, but generally below, unity:

$$\tilde{n} = 1 - \delta + i\beta. \quad (9)$$

The single-interface normal-incidence Fresnel reflectance is too small for a high-throughput projection system. A periodic stack of materials with different complex optical constants allows the small reflected amplitudes from many interfaces to interfere constructively. Mo provides relatively strong optical contrast, while Si provides the spacer layer and comparatively favorable absorption properties in the relevant band.

6.2 Approximate Bragg condition

Let d be one multilayer period and let θ be the ray angle measured from the *surface normal*. The approximate first-order condition is

$$m\lambda \simeq 2d \cos \theta, \quad m = 1, 2, \dots \quad (10)$$

At nearly normal incidence and $m = 1$,

$$d \simeq \frac{\lambda}{2} = \frac{13.5 \text{ nm}}{2} = 6.75 \text{ nm}. \quad (11)$$

The optimum physical design is usually closer to about 6.9–7.1 nm, because Eq. (10) omits the complex-index phase shifts, absorption, interface diffusion, roughness, barrier layers, and capping layer. If the angle is instead defined from the mirror surface, the same relation is written $m\lambda \simeq 2d \sin \alpha$; confusing these two angle conventions causes an apparent sine/cosine discrepancy.

Differentiating Eq. (10) at fixed wavelength gives the first-order angular compensation rule

$$\frac{dd}{d} \simeq \tan \theta d\theta. \quad (12)$$

Thus, a curved mirror on which $\theta = \theta(x, y)$ generally benefits from $d = d(x, y)$. A rigorous coating design replaces Eq. (10) with a transfer-matrix calculation using wavelength- and polarization-dependent complex refractive indices.

6.3 Representative conventional multilayer data

Fraunhofer IWS reports conventional EUV multilayers with approximately

$$t_{\text{Mo}} \sim 3 \text{ nm}, \quad t_{\text{Si}} \sim 4 \text{ nm}, \quad (13)$$

and the measured fabrication/performance ranges shown in Table 6 [10]. These are credible experimental reference values, but they are not a disclosure of the current ZEISS production recipe.

Table 6: Representative experimental Mo/Si multilayer data reported by Fraunhofer IWS [10].

Quantity	Reported value
Approximate Mo thickness	3 nm
Approximate Si thickness	4 nm
Layer-thickness uniformity	$\geq 99.9\%$
Run-to-run reproducibility	99.8–99.9 %
Multilayer microroughness (RMS)	0.15–0.25 nm
EUV reflectance	69.5–70.1 %
Reflectance test condition	$\lambda = 13.5 \text{ nm}, \alpha = 1.5^\circ$
Typical compressive film stress before mitigation	approximately -390 – 420 MPa

The same source notes that curved spherical, aspheric, convex, and concave substrates require controlled one- or two-dimensional thickness gradients. It also emphasizes multilayer stress: even when reflectance is high, coating stress can deform the substrate and thereby create optical figure error.

6.4 Exact interface-engineered stack reported by Bajt et al.

The Mo-on-Si and Si-on-Mo interfaces are not symmetric. Interdiffusion forms molybdenum silicide transition layers, reducing the complex-index contrast and changing the effective period. Bajt *et al.* inserted ultrathin B_4C diffusion barriers and reported the following deposited design for operation near 13.5 nm [9]:

$$\left[\text{Si}(4.37 \text{ nm})/\text{B}_4\text{C}(0.40 \text{ nm})/\text{Mo}(2.25 \text{ nm})/\text{B}_4\text{C}(0.25 \text{ nm}) \right]_{50} + \text{Si}(4.2 \text{ nm}). \quad (14)$$

The two barrier thicknesses are intentionally different because the two interface directions intermix differently. The 0.40 nm barrier is placed at the Mo-on-Si interface, for which silicide formation is stronger, while the 0.25 nm barrier is used at the Si-on-Mo interface.

The deposited layer thicknesses in one period sum to

$$d_{\text{deposited}} = 4.37 + 0.40 + 2.25 + 0.25 = 7.27 \text{ nm}. \quad (15)$$

The same paper reports an interface-engineered period contraction of approximately 0.36 nm per bilayer, giving an effective period near 6.91 nm. This reconciles the apparently large deposited sum with the expected period near 7 nm.

For the 50-period sample, the measured result was

$$R_{\max} = 70.0\% \quad \text{at} \quad \lambda = 13.5 \text{ nm}, \quad \Delta\lambda_{\text{FWHM}} = 0.545 \text{ nm}, \quad (16)$$

measured 5° off normal. The contraction decreased from about 0.52 nm per period for conventional Mo/Si to about 0.36 nm per period with the Mo-on-Si barrier, a reduction of roughly 30 %. The paper also reports that each additional 0.1 nm of high-spatial-frequency substrate roughness can reduce peak reflectance by about two absolute percentage points, motivating substrate roughness near or below 0.1 nm RMS for the highest-reflectance optics [9].

6.5 Capping-layer experiments

A cap must protect the stack against oxidation and contamination without introducing excessive EUV absorption. Bajt *et al.* compared several caps on a separate standard 40-bilayer Mo/Si stack. The most oxidation-resistant configuration tested in that study terminated the stack with

$$\boxed{\text{Si}(2.0 \text{ nm})/\text{B}_4\text{C}(2.0 \text{ nm})/\text{Ru}(2.3 \text{ nm})}. \quad (17)$$

The B_4C separated Ru from the top Si layer and suppressed their interdiffusion. Table 7 records the experimental examples. These were accelerated electron-beam/water-vapor tests from a 2002 study, not present-day scanner lifetime qualifications.

Table 7: Selected capping-layer samples reported by Bajt *et al.* [9]. Values refer to a standard 40-period Mo/Si base stack.

Termination/cap	Initial reflectance	Reported behavior
4.2 nm Si	66.6 % at 13.4 nm	Decreased to 64.9 % after exposure
Top Si + 2.3 nm Ru	65.5 % at 13.45 nm	Ru–Si interreaction observed; exposed area decreased to 64.53 %
2.0 nm Si + 2.0 nm B_4C + 2.3 nm Ru	65.88 % at 13.31 nm	Best oxidation protection among tested designs; no measured loss in the exposed area in that test

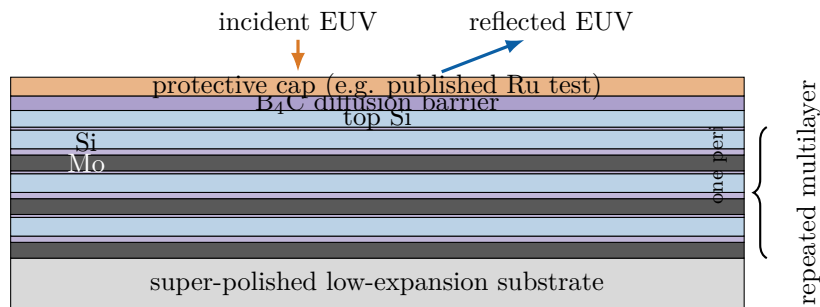


Figure 2: Conceptual multilayer cross-section. Only three periods are drawn; a practical mirror contains tens of periods. Layer heights are schematic and not to scale.

7 Why each mirror cannot use one universal coating

For the chief ray alone, the patent example spans AOIs from 3.1° to 14.6° ; marginal rays at a single surface span a further angular range. Since the reflectance amplitude and phase depend on λ , incidence angle, polarization, and multilayer design,

$$r = r(\lambda, \theta, \text{pol}; d, \Gamma, N, \sigma, \dots), \quad (18)$$

where $\Gamma = t_{\text{Mo}}/d$, N is period count, and σ represents interfacial roughness/diffusion. The optical designer must therefore optimize the coating together with the geometry.

Important consequences are:

1. different mirrors may require different mean periods and Mo fractions;
2. a single mirror may require a two-dimensional period map $d(x, y)$;
3. coating-induced phase must be included in the wavefront budget;
4. polarization splitting at larger angles must be controlled;
5. barrier and cap choices must balance peak reflectance, spectral width, chemical stability, stress, and lifetime.

The Fraunhofer results explicitly document one- and two-dimensional gradient-deposition methods on curved optics [10]. The Zeiss patent demonstrates a complementary geometric strategy: choose mirror pairs whose coating-induced phase variations cancel to first order [7].

8 High-NA optical consequences

Raising the image-side numerical aperture from 0.33 to 0.55 expands the ray cones and therefore increases both mean incidence angles and the angular spread on the wafer-side mirrors. The public ZEISS/ASML High-NA paper describes three central design responses [6].

8.1 Anamorphic magnification

High-NA uses unequal reductions in the two orthogonal directions, conventionally summarized as

$$M_x = \frac{1}{4}, \quad M_y = \frac{1}{8}, \quad (19)$$

with the direction labels depending on the chosen scan-coordinate convention. This preserves the established six-inch mask infrastructure while avoiding overlap of incoming and reflected light cones at the mask. The consequence is a wafer field of $26 \times 16.5 \text{ mm}^2$, half the area of the standard NXE field [6, 3]. A peer-reviewed representative optical design with $M_x = 1/4$, $M_y = 1/8$, $\text{NA} = 0.5$, and the same field dimensions is reported by Liu *et al.* [11]; it should not be interpreted as the proprietary EXE prescription.

8.2 Central obscuration and the final mirror

The High-NA design introduces a hole in the last projection mirror. This lowers otherwise unfavorable angles and angular spread on the last two mirrors, improving coating transmission. The hole appears as a central pupil obscuration. Some diffraction-order information is blocked, so the illuminator and source-mask optimization must explicitly account for the obscured pupil. Partial dark-field illumination can deliberately place the zeroth diffraction order in the obscuration while transmitting useful higher orders, at the cost of exposure-dose and process-window tradeoffs [6].

8.3 Larger and more extreme aspheres

The broader ray cones require larger mirrors and stronger aspheric departures. ZEISS reports that High-NA mirrors are about twice as large and ten times as heavy as established EUV mirrors, and that new vacuum metrology and automated handling were required [5]. The Wischmeier paper describes grinding, polishing, ion-beam figuring, coating, cleaning, figure metrology, and roughness metrology as an iterative manufacturing loop [6].

9 Throughput consequences of finite mirror reflectance

For N reflections with individual reflectances R_i , the purely reflective throughput is

$$T_{\text{refl}} = \prod_{i=1}^N R_i. \quad (20)$$

If every projection mirror had $R = 0.70$, six imaging reflections alone would transmit

$$T_6 = (0.70)^6 = 0.11765 \approx 11.8 \%. \quad (21)$$

For ten equivalent reflections,

$$T_{10} = (0.70)^{10} = 0.02825 \approx 2.8 \%. \quad (22)$$

These are illustrative products, not measured end-to-end scanner efficiencies. A complete power budget must also include source bandwidth, collector efficiency, geometrical acceptance, spectral filters, illumination pupil fill, mask reflectance and absorber loss, obscuration, flare, contamination, and resist absorption.

The leverage of a small absolute reflectance improvement follows from

$$\frac{T_{\text{new}}}{T_{\text{old}}} = \left(\frac{R_{\text{new}}}{R_{\text{old}}} \right)^N. \quad (23)$$

For ten reflections, increasing each mirror from 0.68 to 0.70 yields

$$\left(\frac{0.70}{0.68} \right)^{10} \approx 1.34, \quad (24)$$

or roughly 34% more purely reflective throughput. This exponential multiplication is why interface sharpness, roughness, cap absorption, and contamination control matter at the system level.

10 What is public and what remains proprietary

Table 8: Disclosure matrix for EUV scanner mirror specifications.

Parameter	Public status	Best defensible statement
EUV wavelength	Official	13.5 nm for production NXE/EXE platforms
NXE projection NA	Official	0.33
EXE projection NA	Official	0.55
NXE maximum wafer field	Official	$26 \times 33 \text{ mm}^2$
EXE wafer field	Official/Paper	half field; $26 \times 16.5 \text{ mm}^2$ stated in ZEISS/ASML literature
EXE reduction	Paper	anamorphic, commonly $4 \times / 8 \times$ in orthogonal directions
Established projection mirror count	Official/Paper	six imaging mirrors in the standard-NA Starlith architecture
High-NA illuminator count	Paper	optimized three-mirror illuminator with actuated field facets

Parameter	Public status	Best defensible statement
High-NA final-mirror hole	Paper	creates central obscuration and lowers angle/spread on the last two mirrors
Material-layer count per production mirror	Official	more than 100 material layers
Basic multilayer family	established literature	Mo/Si family optimized for 13.5 nm; may include barriers and caps
Representative Mo thickness	research value	approximately 2.25–3 nm in cited experimental designs
Representative Si thickness	research value	approximately 4–4.4 nm in cited experimental designs
Representative period	research/patent value	approximately 6.9–7.1 nm; one patent example uses 7.04 nm
Representative peak reflectance	research value	approximately 69–71 %, depending on wavelength, angle, polarization, substrate, and stack
Modern NXE M1–M6 radii	Not disclosed	do not substitute the NA = 0.22 patent embodiment
Modern NXE/EXE asphere or freeform coefficients	Not disclosed	proprietary
Exact production Mo and Si thicknesses	Not disclosed	likely mirror- and position-dependent; no authoritative public recipe identified
Exact lateral grading $d(x, y)$	Not disclosed	coating gradients are publicly established, but production maps are proprietary
Current barrier/cap composition and thickness	Not disclosed	published B ₄ C/Ru samples are research references, not proof of current production composition
Modern clear apertures and substrate dimensions	mostly Not disclosed	only limited system-level size comparisons and historic source dimensions are public

The scientifically defensible compact summary is therefore

$$\boxed{d_{\text{Mo/Si}} \sim 7 \text{ nm}, \quad t_{\text{Mo}} \sim 2.25\text{--}3 \text{ nm}, \quad t_{\text{Si}} \sim 4\text{--}4.4 \text{ nm},} \quad (25)$$

with roughly 40–50 or more periods in representative public designs, potentially supplemented by interface barriers, caps, and lateral grading. This is a public design envelope, not a current ASML/ZEISS bill of materials.

11 Recommended hierarchy for simulation

For a physically meaningful open simulation, the information should be used in layers rather than mixed into a fictitious “exact ASML” model:

1. **System architecture model.** Use the collector–illuminator–mask–six-mirror-objective chain and the official values $\lambda = 13.5 \text{ nm}$, $\text{NA} = 0.33$ or 0.55 , and the appropriate field size.
2. **Reconstructable geometric reference.** Implement the $\text{NA} = 0.22$ Zeiss patent prescription in a ray tracer, preserving its sign convention, intermediate image, stop, and aspheric coefficients. Label it explicitly as US 2007/0070322 A1.

3. **Wave-optical coating model.** Use a transfer matrix with complex $n(\lambda)$ for Mo, Si, barriers, and cap. Begin with 40 periods at 7.04 nm or the exact Bajt stack, then vary d , Γ , N , interface width, roughness, and cap.
4. **Angle-coupled coating map.** Trace the local AOI distribution on each mirror and optimize $d(x, y)$ jointly for amplitude and phase. Include separate s - and p -polarized coefficients.
5. **Power budget.** Propagate measured or computed $R_i(\lambda, \theta, \text{pol})$ through Eq. (20), adding collector, mask, pupil-fill, obscuration, contamination, and geometric factors separately.
6. **Uncertainty statement.** Report which values are official, which come from the patent, and which come from research coatings. Do not optimize a surrogate model and then claim that the result is an improvement over an undisclosed production stack.

This hierarchy is especially important for a throughput-improvement project. A change that raises normal-incidence peak reflectance may still reduce integrated scanner throughput if it narrows the spectral band, increases polarization splitting, introduces coating phase error, raises stress, worsens lifetime, or performs poorly over the real AOI distribution.

12 Conclusion

The public record supports a precise structural picture but not a complete reverse engineering of current ASML/ZEISS optics. The source is imaged to an intermediate focus by a large, graded collector; faceted illumination optics synthesize the field and pupil; a reflective mask produces diffraction orders; and a multi-mirror projection objective forms the reduced wafer image. Standard-NA Starlith optics use six imaging mirrors. High-NA adds anamorphic imaging, a half-size field, a three-mirror illuminator, larger and more extreme aspheres, and a hole in the last projection mirror that produces a central obscuration.

At the coating level, public sources strongly support a Mo/Si period near 7 nm, individual layers of order 3 nm Mo and 4 nm Si, peak research reflectance near 70 %, sub-nanometre interfacial roughness, and the importance of barriers, caps, stress control, and lateral grading. The exact Bajt interface-engineered sample and the full Zeiss six-mirror patent prescription provide unusually detailed, reconstructable reference cases. They should be treated as research and patent benchmarks, respectively—not as leaked production specifications.

A Sequential data for the preferred patent embodiment

Table 9 reproduces the signed sequential distances in the patent. In a reflective sequential model the negative distances arise from reversal of propagation direction after reflection. Software-specific coordinate breaks, mirror apertures, and off-axis used regions still need to be implemented consistently; the table alone should not be interpreted as a mechanical layout.

B Minimum metadata to attach to any quoted mirror value

Any future mirror database should record at least:

$$\left\{ \begin{array}{l} \text{source and evidence class,} \\ \text{scanner or reference design,} \\ \lambda, \text{ NA, field point and pupil point,} \\ \text{angle convention and polarization,} \\ R, K, \{A, B, C, D, E\}, \text{ used aperture,} \\ d(x, y), \Gamma(x, y), N, \text{ barriers and cap,} \\ \text{roughness, diffusion width, stress, age, and test environment.} \end{array} \right. \quad (26)$$

Table 9: Sequential prescription from Table 1 of US 2007/0070322 A1 [7].

Surface	Vertex radius (mm)	Signed distance to next surface (mm)	Type
Object (mask)	∞	723.7539	object plane
M_1	-643.5439	-192.7897	reflective asphere
Aperture stop	∞	-99.4340	stop
M_2	-2742.4833	363.1533	reflective asphere
Intermediate image	∞	343.7066	image plane
M_3	-2850.8680	-644.4521	reflective asphere
M_4	1471.8440	971.0620	reflective asphere
M_5	252.1454	-475.4847	reflective asphere
M_6	558.1165	510.4847	reflective asphere
Image (wafer)	∞		image plane

Without this metadata, two apparently contradictory thickness or reflectance values may simply refer to different angles, wavelengths, polarizations, locations, or interface models.

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